

21 June P2 + ECAA-predecessor recall sheet

Local-only recall sheet built from the frozen 21 June final revision guide.

It is a teaching aid, not an exam question paper; the status lines below record which items were completed live and which are official-reference completion.

Scope boundary

ECAA specimen Q15 and Q16 are historical UCLES ECAA 2020 material used for ESAT-style

practice. They are NOT current official ESAT questions. Current ESAT is on-screen and

calculator-free; formula-booklet availability for it is not independently verified from the cited ESAT source set, so no current-ESAT formula-booklet claim is asserted here.

ECAA-predecessor warm-ups (historical practice, not current ESAT)

ECAA Q15 - rigid transformations (answer B)

- Square vertices (+/-1,+/-1); rotate 90 deg cw then reflect in $y=x$.
- Track every vertex (the lesson method): P(1,1) Q(-1,1) R(-1,-1) S(1,-1).
- Net effect is reflection in the y-axis; it is its own inverse, so answer B.
- The matrix composition is a Derived verification check, NOT the recorded lesson method. Under exam conditions show the vertex table.

ECAA Q16 - conditional probability (answer E, 2/3)

- Bus on time 0.6 (late 0.4). Catch train: 0.8 if on time, 0.6 if late.
 - Tree: on time then catch = 0.48; late then catch = 0.24; total catch = 0.72.
 - $P(\text{on time} \mid \text{catches}) = 0.48 / 0.72 = 2/3$. Do not assume independence.
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P2 question targets

Q1 - arithmetic series (2 + 2 marks)

- $a = 78, d = -3.$
- $u_{18} = 78 + 17(-3) = 27.$
- $S_{100} = 50 \times [2(78) + 99(-3)] = 50 \times (-141) = -7050.$

Q2 - binomial expansion (6 marks)

- $(1+px)^{10} = 1 + 10px + 45p^2 x^2 + 120p^3 x^3 + \dots$
- $10p = 15 \Rightarrow p = 3/2.$
- $q = 45p^2 = 405/4 = 101.25; r = 120p^3 = 405.$
- The unfinished submitted state earned 0/6; simplify before matching coefficients.

Q3 - logs and trapezium (2 + 3 + 2 marks)

- (a) $3 \times 2^{(-k)} = 300000 \Rightarrow 2^{(-k)} = 100000 \Rightarrow k = -\log(100000)/\log(2)$ approx -16.61.
- (b)(i) Six tabulated values $\Rightarrow$ FIVE strips; $h = (7 - (-0.5))/5 = 1.5.$
- (b)(i) Trapezium estimate approx 6.63.
- (b)(ii) Follow-through approx 4.42.
- The mis-count (six values = five strips) cost three marks.

Q4 - division, factor, proof, decreasing (4 + 5 + 3 marks)

- (a) Divide $f(x)=4x^3+13x^2-10x+8$ by $x-2$: quotient $4x^2+21x+32$, remainder 72.
- (b)(i) $f(-4)=0 \Rightarrow (x+4)$ a factor.
- (b)(ii) $f(x)=(x+4)(4x^2-3x+2)$, discriminant $-23 < 0 \Rightarrow$ exactly one real root $x=-4.$
- (c) $f'(x)=2(3x-1)(2x+5) < 0 \Rightarrow -5/2 < x < 1/3.$

Q5 - trig equations (5 + 4 marks)

- (i) Requires $\cos(\theta) \neq 0.$ $4 \tan(\theta)+5 \sin(\theta)=0 \Rightarrow \sin(\theta)(4+5 \cos(\theta))=0.$
- (i) $\theta = 143.1, 180, 216.9, 360$ degrees.
- (ii) Requires $\sin(x) \neq 0$ and $\cos(x) \neq 0.$ Multiply by $\sin(x)\cos(x).$
- (ii) $\sin^2 x + \cos^2 x = 5 \sin x \Rightarrow \sin x = 0.2 \Rightarrow x = 0.201, 2.94$ rad (3 s.f.).

Q6 - circles (5 + 4 marks)

- (a) Diameter A(-2,18) B(14,6): centre (6,12), $r = 10 \Rightarrow (x-6)^2+(y-12)^2 = 100.$
- (b) Origin-to-centre distance = $\sqrt{180} = 6\sqrt{5}.$
- (b) C2: $x^2 + y^2 = 280 - 120\sqrt{5}$ OR $x^2 + y^2 = 280 + 120\sqrt{5}$ (two circles touch C1).

Q7 - differentiation and area (2 + 2 + 6 marks)

- (a)(b) $\frac{dy}{dx} = 3x^2 - 10x^{3/2} - \frac{k}{2}x^{-1/2} + 28$; at $x=9$ gives $k=6$.
 - (c) Exact shaded area = $6939/28$ approx 247.821 $\Rightarrow$ 248 (3 s.f.).
 - A negative area flags a sign slip (curve above line here, so the integral is positive).
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Deferred (official-reference results only, not taught)

Q8 (geometric sum + evaporation model)

- Geometric sum $S_n = a(1-r^n)/(1-r)$; $150(0.92)^6$ approx 91 L; 40-year total approx 1664 L.
- Write approx, not equals.

Q9 (log equation + proof $a = b/(b-1)$, $b > 1$)

- $(4x+5)^2 = 9(x+3) \Rightarrow 16x^2+31x-2 = 0 \Rightarrow x = 1/16$ only (reject $x = -2$; $4x+5 > 0$ fails).
- Domain rejection is a real mark, not optional.

Attempt Q8 and Q9 unaided before a later revisit; the boxes above are reference results only.